RAFAEL RIBEIRO TONHOLO

SENIOR ANDROID ENGINEER | KOTLIN, COMPOSE & MOBILE PLATFORM

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Senior Android engineer with 11+ years delivering and modernizing production mobile apps, including Thunderbird for Android (1M+ monthly active users). Deep experience in Kotlin, Jetpack Compose, Coroutines/Flow, MVVM/MVI, Gradle tooling, app performance, and automated testing. Builds open-source developer tools, collaborates across platforms, and uses AI coding assistants with critical review.

WORK EXPERIENCE

Senior Mobile Engineer | Thunderbird Apr 2025 – Oct 2026

Remote, Canada

- Modernized Thunderbird for Android's message list (1M+ monthly active users), transitioning legacy UI toward Jetpack Compose with an MVI state machine and compatibility bridges for existing interactions.
- Designed reusable system and in-app notification infrastructure with Compose banners, snackbars, and dialogs, centralizing delivery, dismissal, and priority handling across product features.
- Introduced Android startup benchmarks and Baseline Profile generation for Thunderbird and K-9 Mail, measuring time to initial/full display across compilation modes.
- Built a declarative feature-flag platform with a Gradle plugin, JSON-schema validation, generated keys, runtime overrides, and remote catalogs to support configurable releases.
- Developed Thundermail account onboarding with Kotlin and Compose, integrating OAuth sign-in, QR-code entry points, feature modules, and dependency injection across app variants.
- Diagnosed and fixed production crashes, including a database migration that risked local data loss and a message-composer crash triggered by screen rotation.
- Built a Kotlin compiler plugin for PII-safe logging and automated PR policy checks; used AI-assisted coding workflows for compiler/CI tasks with tests and code review.

Selected merged contributions: notifications · feature flags · Compose migration · PII-safe logging

Senior Mobile Engineer | REDspace Mar 2022 – Apr 2025

Halifax, Canada

- Delivered an accessible Android mobile and TV streaming app in approximately 6 months using Kotlin, Jetpack Compose, MVVM, and Media3 offline playback.
- Re-architected Bleacher Report ad placements with server-driven rules, eliminating concurrent list-update crashes and enabling new slots without app releases.
- Designed shared Kotlin Multiplatform analytics for Android and iOS, reusing business logic while retaining platform-native application architectures.
- Introduced Jetpack Compose to legacy Android codebases and helped teams adopt reusable UI components and modern architecture.

Senior Android Engineer | Pinterest Jul 2021 – Mar 2022

Remote, USA

- Reduced Android incremental build time from 15 to 7 minutes by modularizing a large codebase into 30 feature modules.
- Built Hilt dependency management for Android Dynamic Feature Modules, improving stability of dynamically delivered features.

Principal Software Engineer | Questrade Financial Group Oct 2019 – Jul 2021
Remote, Canada

- Provided technical leadership for 3 mobile teams preparing 2 React Native apps for Android and iOS, guiding mobile architecture, API integration, and delivery.
- Led migration from Apache Cordova toward React Native and mentored engineers on cross-platform implementation and engineering practices.

Senior Software Engineer | Questrade Financial Group Aug 2017 – Sep 2019
Remote, Canada

- Improved a financial mobile app serving 100,000+ daily users; its app-store rating increased from 1.5 to 3.3 stars.
- Delivered native Android and iOS authentication enhancements, including biometrics, supported by automated tests and CI/CD pipelines.

Android Engineer | Axxiom Jun 2015 – Aug 2017
Belo Horizonte, Brazil

- Modernized a Windows CE utility field-service app for Android with Google Maps, improving support for outage-repair workflows
- Built offline-first synchronization with SQLite and AIDL inter-app communication for field work over unreliable satellite links

SKILLS

Languages: Kotlin, Java, SQL, JavaScript, Swift

Android: Jetpack Compose, Coroutines, Flow, Lifecycle, Navigation, Room, Media3, Retrofit, Hilt, Dagger, Koin

Architecture & tooling: MVVM, MVI, Kotlin Multiplatform, modularization, REST APIs, Gradle plugins, Kotlin compiler plugins

Testing, delivery & AI: JUnit, Robolectric, Espresso, Compose UI tests, Macrobenchmark, Baseline Profiles, GitHub Actions, GitLab CI, Bitrise, Claude Code, Codex, MCP

PROJECTS

svg-to-compose | Open-Source Developer Tool
Website: svgtocompose.com • GitHub: rafaeltonholo/svg-to-compose • Install guide

- Published a Kotlin Multiplatform CLI and incremental Gradle plugin converting SVG/Android vector assets into Jetpack Compose ImageVector code, distributed via Maven Central.
- Built a custom parser for complex vector geometry and an interactive web playground with examples and documentation at svgtocompose.com.

EDUCATION

UNA University Center | Postgraduate Specialization in Mobile Application Development, 2016
PUC Minas | Technologist Diploma in Digital Games, 2014
WES Canadian equivalencies: 1-year postgraduate diploma and 3-year diploma, respectively